RECOMMENDATION SYSTEMS

AI & Machine Learning | Spring 2026

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Agenda

Topics Covered

Basic Introduction

In depth discussion of algorithms

Code exercise

Conclusion & Questions

NETFLIX

TV Shows Movies Recently Added My List

N SERIES

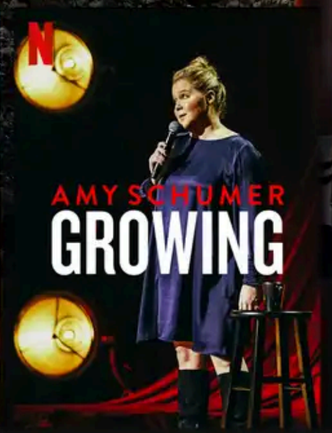
STRANGER THINGS

When a young boy vanishes, a small town uncovers a mystery involving secret experiments, terrifying supernatural forces and a strange little girl.

Play

+ My List

Popular on Netflix



Introduction

- AI systems play a huge role in recommendation systems on streaming platforms like Netflix.
- Instead of users searching through thousands of movies and shows, AI analyzes things like what you watch, how long you watch it, and what you like to recommend content you'll enjoy.
- These systems use machine learning to find patterns in your behavior and compare it with other users to make accurate suggestions.
- Because of this, most of the content people watch on Netflix actually comes from recommendations, not searches . This makes the platform more personalized and keeps users engaged.

Types of Recommendation Systems

- Content based filtering
- Collaborative filtering

Collaborative Filtering

- Focuses on user similarity to recommend items
- Uses user behavior to recommend content
- Finds people with similar tastes
- Suggests movies/shows liked by similar users
- Based on the idea: “People like you also liked...”

Netflix Ratings

USERS	GINNY AND GEORGIA	NEVER HAVE I EVER	YOU	STRANGER THINGS
ISABELLA	3	?	5	3.5
CHIDI	4	4	2	?
GRACE	4	?	2.5	4

Chidi and Grace have similar ratings. Based on this, Chidi would likely enjoy Stranger things and Grace would likely enjoy Never Have I ever.
(This is not true- Chidi hates scary shows!)

Problems with Collaborative Filtering

Cold Start Problem

New users/items have no interaction data

Data Sparsity

Most users interact with very few items

Scalability Challenges

Difficult to handle large numbers of users and items

Limited Early Recommendations

System needs enough data before it works well

Content based filtering

- Incorporates machine learning algorithms and data science techniques to recommend new items and answer queries.
  - The system compares:
    a. User profile - user preferences and behavior
    b. Item profile - movies according to genre, release date.
- This method will compare both profiles and predict user-item interaction and recommend items accordingly.
- A major algorithm used is **Cosine Similarity.**

Ph.D, Jacob Murel, and Eda Kavlakoglu. “Content-Based Filtering.” *Ibm.Com*, 21 Mar. 2024.

Cosine Similarity

- Measures similarity between two items
- Compares items as vectors of features
- Value ranges from:
  - 1 → very similar
  - 0 → not similar

$$\cos\theta = \frac{AB}{|A|\,|B|}$$

In Class Participation Activity

Want to try it yourself?
Access our code here:

https://github.com/chidiiiiiii/AI_MachineLearningClass/tree/main/Final%20presentation

Problems with Content-Based Filtering

Limited Recommendations

Only suggests items similar to what the user already likes

Over-Specialization

Lacks variety and discovery of new content

Feature Dependency

Depends heavily on quality of item descriptions (e.g., synopsis)

No Community Insight

Does not consider other users' preferences

Conclusion

AI powers modern recommendation systems

Collaborative filtering - “people like you”

Content-based filtering - “items like this”

Both methods have limitations

Thank You!
Any Questions?